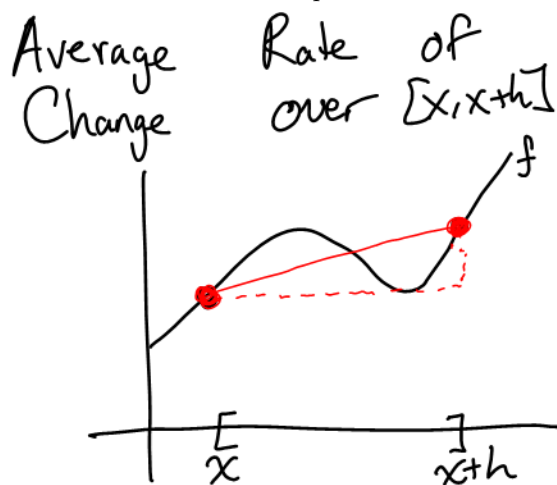
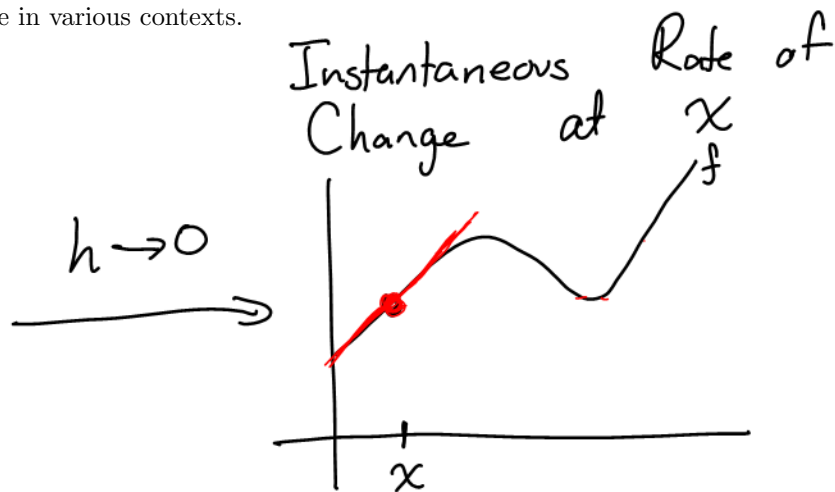


Lecture 7: Instantaneous Rates of Change

GOAL: Interpret the role of the derivative in various contexts.



$$\text{Avg Change} = \frac{f(x+h) - f(x)}{h}$$



$$\text{Inst. Change} = f'(x)$$

Physics

Velocity: $\text{Velocity} = \frac{\text{Change in Position}}{\text{Time}} = \frac{\Delta s(t)}{\Delta t}$

If $s(t)$ represents the position of an object at time t , then the velocity $v(t)$ is given by

$$v(t) \stackrel{\text{def}}{=} \frac{ds}{dt}$$

* NOTE: Unless otherwise stated, t is time in seconds

Example 1

The position of a particle (in meters) moving in a straight line is given by

$$s(t) = 7 \cos t + t^2$$

What is the velocity?

$$\begin{aligned}
 v(t) = s'(t) &= 7 [\cos t]' + [t^2]' \\
 \frac{ds}{dt} &= 7(-\sin t) + 2t \\
 &= [2t - 7 \sin t] \frac{m}{s}
 \end{aligned}$$

★ NOTE: Unit of Derivative = $\frac{\text{Unit of dependent Var}}{\text{Unit of Independent Var}}$

Example 2

A ball is tossed straight into the air. The height from the ground (in feet) can be modeled via

$$s(t) = 80t - \underbrace{16}_{\frac{1}{2}g} t^2$$

$$\begin{aligned}
 &-9.81 \frac{m}{s^2} \\
 &\approx -32 \frac{ft}{s^2}
 \end{aligned}$$

@ Velocity?

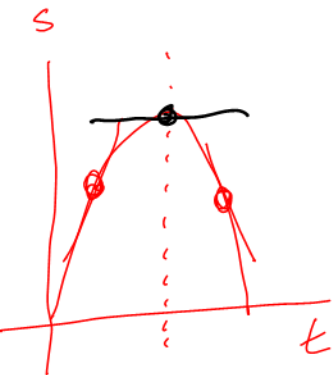
$$v(t) = s'(t) = [80 - 32t] \frac{ft}{s}$$

⑤ When $t=3$, is the ball going up or down?

$$v(3) = 80 - 32 \cdot 3 = 80 - 96 = -16 \frac{ft}{s}$$

$v(3) < 0 \Rightarrow$ height decreasing

$\Rightarrow$ Ball is going down



⑥ When will the ball reach its peak?

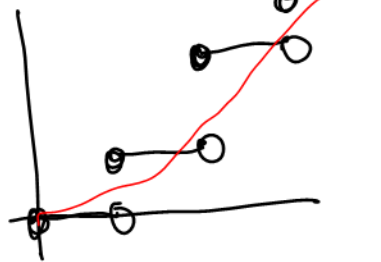
$$v(t) = 80 - 32t \stackrel{\text{set}}{=} 0$$

$$80 = 32t$$

$$t = \frac{80}{32} = 2.5 \text{ seconds}$$

Biology

Population Growth



$P(t)$ represents the population at time t

$$P(t) = P_0 \cdot e^{kt}$$

P_0 Initial Pop. k Some constant

$\frac{dP}{dt}$ is called the population growth; measures how fast the population grows or shrinks.

Example 3 The population in a city since 2000 can be modeled via the equation

$$P(t) = 100t^2 - 600t + 10000$$

@ At what rate is the population changing?

$$P'(t) = [200t - 600] \frac{\text{people}}{\text{yr}}$$

⑤ What year is the population increasing at a rate of 1000 people/yr?

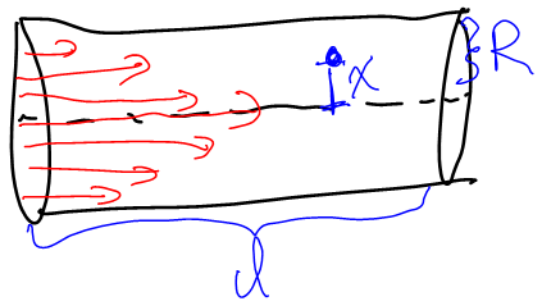
$$P'(t) = 200t - 600 \stackrel{\text{set}}{=} 1000$$

$$200t = 1600$$

$$t = 8$$

$\Rightarrow$ In the year 2008

Blood Flow



The velocity of blood is modeled by Pressure Difference

$$V(x) = \frac{P}{4\eta l} (R^2 - x^2)$$

η is viscosity, l is length

$\frac{dV}{dx}$ is called the velocity gradient.

Example In a small artery, the velocity of the blood is modeled

$$V(x) = (1.85 \times 10^4) (6.4 \times 10^{-5} - x^2)$$

x is in centimeters.

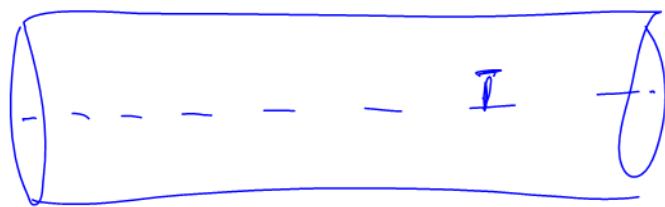
$$\begin{aligned} \textcircled{a} \quad \frac{dV}{dx} &= (1.85 \times 10^4) [6.4 \times 10^{-5} - x^2]^{-2x} \\ &= [-2(1.85 \times 10^4)x] \frac{\text{cm/s}}{\text{cm}} \end{aligned}$$

② Evaluate at $x = 0.002 \text{ cm} = 2 \times 10^{-3} \text{ cm}$

$$v'(0.002) = -2(1.85 \times 10^4)(2 \times 10^{-3}) = -74 \frac{\text{cm/s}}{\text{cm}}$$

NOTE: $1 \text{ cm} = 10000 \mu\text{m}$

$$-74 \frac{\text{cm/s}}{\text{cm}} = -74 \frac{\mu\text{m/s}}{\mu\text{m}}$$



So, if we move $1 \mu\text{m}$ away from the center of the artery, the velocity decreases by $74 \frac{\mu\text{m}}{\text{s}}$.

Economics Cost Functions



The cost function $P = C(q)$ represents the costs to produce q units.

$C(0)$ is called the overhead cost.

$\frac{dp}{dq} = \frac{dC}{dq}$ is called the marginal cost.

NOTE: It is very common to represent cost functions as cubic

Example 5

In USD, a company estimates the cost of producing q units as

$$C(q) = 10000 + 5q + 0.01q^3$$

What's the marginal cost at 50 units?

$$\begin{aligned} C'(q) &= 5 + 0.01(3q^2) \\ &= \$ (5 + 0.03q^2) / \text{unit} \end{aligned}$$

$$C'(50) = 5 + 0.03(50)^2 = \$80 / \text{unit}$$

$$\text{NOTE: } C'(50) \approx \frac{C(51) - C(50)}{51 - 50} = \$81.51 / \text{item}$$

Profit

The Revenue Function $R(q)$ is the amount of money made after producing q units (w/o factoring in expenses)

Profit $\stackrel{\text{def}}{=}$ Revenue - Cost

$$P(q) \stackrel{\text{def}}{=} R(q) - C(q)$$

Example 6 $R(q) = q$

$$C(q) = 50 - 6q^2 + 4q^3$$

Find $P'(q)$

$$P'(q) = [R(q) - C(q)]'$$

$$= R'(q) - C'(q)$$

$$= 1 - (-12q + 12q^2)$$

$$= -12q^2 + 12q + 1$$